

- Q.27 Explain Radiation shield and view factor with reference to radiation heat transfer.
- Q.28 Explain with example, each mode of heat transfer.
- Q.29 State and Explain Kirchoff's law with respect to radiation heat transfer.
- Q.30 Differentiate boiling and evaporation.
- Q.31 Explain the convection mode of heat transfer with example.
- Q.32 State Stefan-Boltzmann law.
- Q.33 Write short note on view factor in radiation.
- Q.34 What is the difference between steady-state and unsteady-state heat transfer?
- Q.35 Discuss one dimensional steady state heat conduction through a Plain wall.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Drive the expression for steady – State one dimensional steady state heat conduction through a composite wall. Also mention the assumptions used.
- Q.37 Write short note on any two:
1. Nusselt number & Grash of number
 2. Weni's Displacement law
 3. Kirchhoff's law
- Q.38 Explain critical thickness of insulation with its derivation. Also discuss its significance

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Roll No.

3rd Sem / Chemical Engineering, P & P Sub : Heat Transfer – I

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Driving force for heat transfer is:
- a) Time
 - b) Temperature
 - c) Pressure
 - d) Velocity
- Q.2 The critical thickness of insulation is mainly dependent on:
- a) Material conductivity
 - b) Surface area
 - c) Temperature difference
 - d) Insulation colour
- Q.3 Fourier law of heat conduction is best represented by
- a) $Q = -kA \frac{dt}{dx}$
 - b) $Q = kA \frac{dx}{dt}$
 - c) $Q = -kA$
 - d) $Q = k \frac{dt}{dx}$
- Q.4 Why glass wool is used?
- a) For insulation
 - b) For Conduction
 - c) For Convection
 - d) Radiation

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- Q.5 The body which absorbs all radiations incident upon it is called as
- Black Body
 - White Body
 - Opaque Body
 - Transparent Body
- Q.6 Why is insulation used?
- To permit flow of heat
 - To prevent flow of heat
 - To reduce pressure
 - To increase pressure
- Q.7 Which heat transfer mechanism occurs through the movement of fluid particles?
- Conduction
 - Convection
 - Radiation
 - Both A and C
- Q.8 Which of the following is a method of heat transfer?
- Convection
 - Radiation
 - Conduction
 - All of the mentioned
- Q.9 Ratio of momentum diffusivity of thermal diffusivity is known as
- Reynold's number
 - Prandtl number
 - Nusselt number
 - Peclet number
- Q.10 Which mode of heat transfer does not necessarily require the presence of a medium?
- Conduction
 - Convection
 - Radiation
 - None of these

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Write the S.I unit of thermal conductivity.
- Q.12 Define emissive power.
- Q.13 What is the driving force for heat transfer?
- Q.14 What does the steady state heat transfer means?
- Q.15 What is conductor?
- Q.16 Why negative sign is used in Fourier's law?
- Q.17 Write one example of good insulator.
- Q.18 Write full form of LMTD.
- Q.19 What is black body?
- Q.20 Give an example of natural convection.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Derive the expression for critical thickness of insulation for sphere.
- Q.22 Write applications of dimension less numbers.
- Q.23 Explain Fourier's law of heat conduction.
- Q.24 Explain the concept of heat transfer and its applications in chemical engineering.
- Q.25 Differentiate between free and forced convection.
- Q.26 Explain absorptivity and reflectivity.

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